卷册检索号 SEARCH NO. OF THE VOLUME
MLHVDC-LCS-ECI-DD-CS-0249 30-B003701S-T0249

图纸文件目录（首页）

CATALOG OF DRAWING DOCUMENTS (HEAD PAGE)

拉合尔±660kV换流站 工程 施工图 阶段
±660kV Lahore Converter Station PROJECT Detail Design DESIGN STAGE

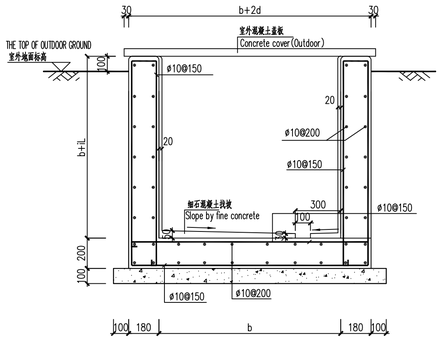
结构 专业 第 02 卷 第 49 册
Structure SPECIALITY VOLUME BOOK

卷册名称 结构通用详图（适用于阀厅、控制楼、换流变基础）
Volume name General layout of structure (applicable to foundations of valve hall, control building and change and rheology building)

批　准 APPROVED BY	2018.02.28
审　核 REVIEWED BY	
设计/勘测 DESIGNED BY/ SURVEYED BY	2018.02.27
图 纸 6 张 DRAWING NOS.	打印件 本 PRINTED COPIES NOS.

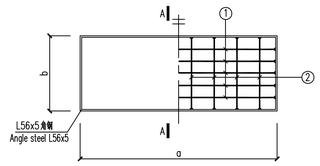
序号 No.	图　号 DRAWING NO.	文 件 名 称 DOCUMENT TITLE	版本 EDITION	张 NOS.	本 NOS.	备注 REMARKS
1	30-B003701S-T0249-001	电缆沟详图(一) Construction details of Cable Trench (I)	A	1		
2	30-B003701S-T0249-002	电缆沟详图(二) Construction details of Cable Trench (II)	A	1		
3	30-B003701S-T0249-003	预埋件总说明(一) General Description of the Embeded Steel (I)	A	1		
4	30-B003701S-T0249-004	预埋件总说明(二) General Description of the Embeded Steel (II)	A	1		
5	30-B003701S-T0249-005	单轨吊总说明 GENERAL DESCRIPTION OF MOTOR HOIST MONORAIL	A	1		
6	30-B003701S-T0249-006	混凝土支承梁与轨道连接图 MONORAIL CONNECTION DETAILS WITH CONCRETE BEAM	A	1		

注:



室外电缆沟截面图

RC. TRENCH (Outdoor)



室外混凝土盖板

Concrete cover(Outdoor)

室外电缆沟盖板设计活荷载 4kN/m².

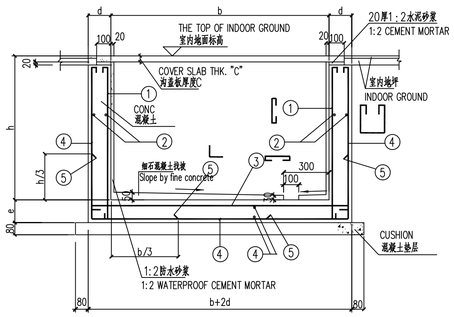
The design live load on trench cover(outdoor) is 4kN/m².

A–A

室外电缆沟盖板一览表

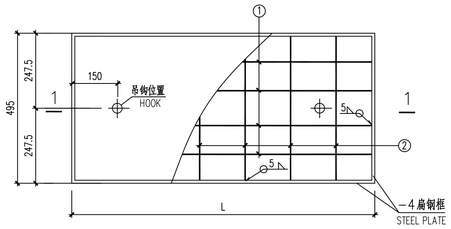
Concrete cover list(Outdoor)

No.	Width	Parameter a	Parameter b	Thickness	①	②



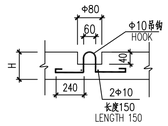
室内钢筋混凝土沟道断面图

RC. TRENCH (INDOOR)



室内混凝土盖板平面图

PLAN OF CONCRETE COVER PLATE(Indoor)



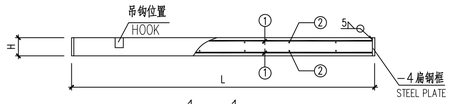
吊钩详图

HOOK DETAIL

室内电缆沟盖板一览表

Concrete cover list(Indoor)

序号 NO.	沟道宽 TRENCH WIDTH b(mm)	盖板编号 COVER PLATE NAME	沟盖板长度 PLATE LENGTH L(mm)	沟盖板厚度 PLATE THICKNESS H(mm)	①	②
1	1000	1NHB10–10	1000	100	2x4Φ10	2x6Φ10
2	1000	1NHB10–11.5	1150	100	2x4Φ10	2x8Φ10
3	1200	1NHB12–12	1200	100	2x4Φ10	2x8Φ10
4	1200	1NHB12–13.5	1350	100	2x5Φ10	2x10Φ10
5	1000	3NHB12–12	1200	100	2x6Φ10	2x8Φ10



1——1

室内钢筋混凝土沟道配筋

REINFORCEMENT FOR TRENCH(INDOOR)

b (mm)	h (mm)	d	e	①	②	③	④	⑤
b≤5400	1000<h≤1200	150	300	ø16@200	Φ10@200	ø16@200	ø16@200	ø16@200

电缆沟说明:

Notes for construction of cable trenches:

1.电缆沟设计使用年限为50年。

1.The design service limit is 50 years.

2.电缆沟均以中心线为定位。

2.All the cable trenches are located by center axis.

3.混凝土强度采用fc'=28Mpa,钢筋ASTM A615 Grade 60, fy = 420MPa。
垫层采用100厚fc'=21Mpa混凝土

3.The strength of concrete used in foundation is fc'=28Mpa,reinforcement ASTM A615 Grade 60, fy = 420MPa.
The strength of concrete used in cushion is fc'=21Mpa.

4.电缆沟及盖板的混凝土保护层厚度如下表:

4.The thickness of concrete protection is as following table:

部位	保护层	
侧壁	内侧	19.1mm
	外侧(室外)	76.2mm
	外侧(室内)	50.0mm
底板	上侧	19.1mm
	外侧(室外)	76.2mm
	外侧(室内)	50.0mm
盖板	下侧	19.1mm

Items	Concrete protection	
Side wall	Interior	19.1mm
	Exterior(Outdoor)	76.2mm
	Exterior(Indoor)	50.0mm
Base slab	Top	19.1mm
	Bottom (Outdoor)	76.2mm
	Bottom (Indoor)	50.0mm
Cover	Bottom	19.1mm

5.电缆沟底纵向排水坡度 i=0.5%,每隔30米左右设一集水坑,集水坑用埋管通向附近窨井。

5.The longitute slope for drainage at bottom is i=0.5%. A sump pit should be set at each 30m long, and connected to the manhole nearby by pipe.

6.电缆沟每隔15米设一道25mm伸缩缝,填充材料采用新型聚氨酯类材料填充,表面材料采用中性硅酮耐候密封胶

6.A 25mm wide joint should be set at each 15m long of cable trench.
The joint should be filled with polyurethane materials inside and finished by neutral silicone weather sealant.

7.外露铁件热镀锌防腐;现场焊接破坏时采取防腐措施,底漆:环氧富锌;中漆:云铁氯化橡胶;面漆:氯化橡胶丙烯酸磁漆。

7.All the exposed steel members should be hot-galvanized. The steel member welded in site should be paint coating for anti-corrosion.
Priming coating: Epoxy zinc rich paint; Second coating: Ferric chloride chlorinated rubber paint; Finishing coating: Chlorinated rubber acrylic enamel paint.

8.电缆沟需采用好土回填,并分层夯实,压实系数不小于0.94。

8.The soil under cable trenches should be good quality soil and tamped by layers, the compacting factor should be more than 0.94.

9.电缆沟压顶梁上铺设30mmx10mm橡胶垫。

9.The 30mmx10mm rubber pad should be equipped on the cappings of trench.

10.地下混凝土要求:
1)水胶比不大于0.45;
2)最大氯离子含量(水泥用量的百分比):0.10;测试方法参《ASTM C 1218M》。
3)水泥型号:采用抗硫酸盐的硅酸盐水泥,水泥型号需满足《ACI 318–14》表19.3.2.1对应于暴露等级S1的水泥种类要求(ASTM C150 或ASTM 595 或ASTM C1157对应的类别);
4)水泥中掺入矿物掺合料(矿物掺合料的掺量、使用方法和耐腐蚀性能按相应产品的使用说明并经验证后确定)。

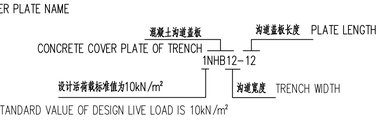
10.Requirements for Underground Concrete:
1) Maximum water–cement ratio is 0.45;
2) Maximum water–soluble chloride ion content is 0.1;Test method refers to "ASTM C 1218M".
3) Type of Concrete: Concrete should be made of sulfate–resistance silica cement.Cement type should fulfill the requirements for concrete used in exposure grade S1, according to《ACI 318–14》table 19.3.2.1.(ASTM C150,ASTM 595 or ASTM C1157 corresponding cement type).
4) Add mineral admixtures (the dosage, using method and anti–corrosion ability should be tested according to the product manual)

11. 吊钩应采用ASTM A615 Grade 40级钢筋制作,严禁使用冷加工钢筋。

11. HOOK SHALL BE MADE BYASTM A615 Grade 40. COLD FORMING REINFORCEMENT BAR SHALL BE FORBIDDEN.

12. 盖板编号说明

12. COVER PLATE NAME



13.与建筑物相连的电缆沟定位供参考,以建筑结构图纸为准。

13.The location of cables which connectes to building is reference only, and it will subject to the structural drawings of building.

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC EQUIPMENT GROUP CORPORATION CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C – L C S – E C I – D D – C S – 0 2 4 9 – 0 0 1 A

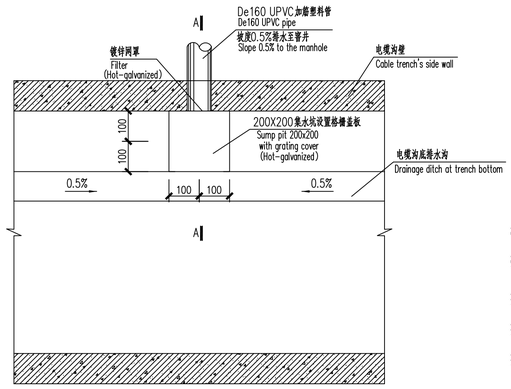
	姓名 Name	日期 Date
设计 Designed	孙垚垚	2018.02.27
校核 Checked	刘婷	2018.02.28
审核 Reviewed	郑湘诚	2018.02.28
批准 Approved	王柳强	2018.02.28

电缆沟详图(一)

Construction details of Cable Trench (Ⅰ)

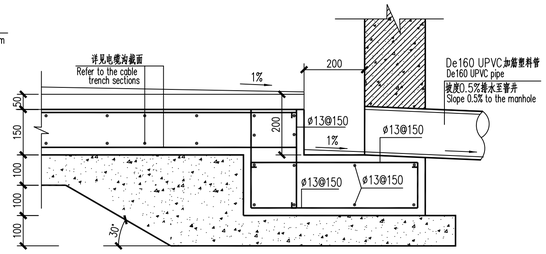
比例 Scale: | 图号 DWG NO.: 30-B003701S-T0249-001 | 版本 REV.: A

A1(841x594)

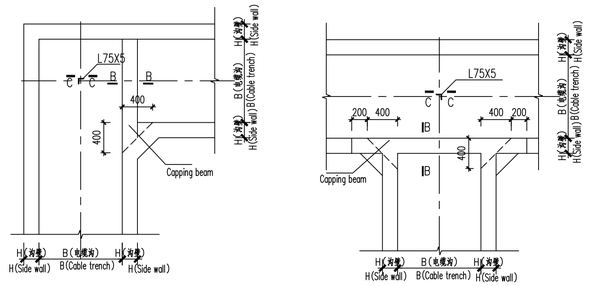


电缆沟集水井详图

Details of sump pit

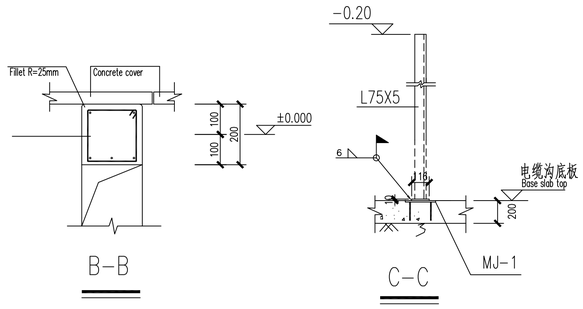


A-A



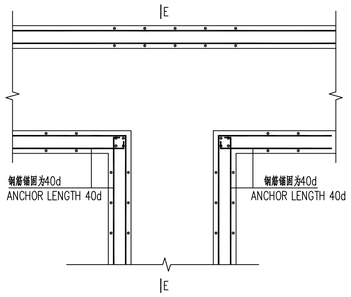
沟道交错布置图(室外)

Cover plate plan for intersection trench(Outdoor)



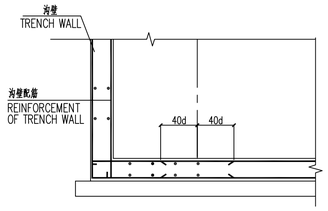
B-B

C-C

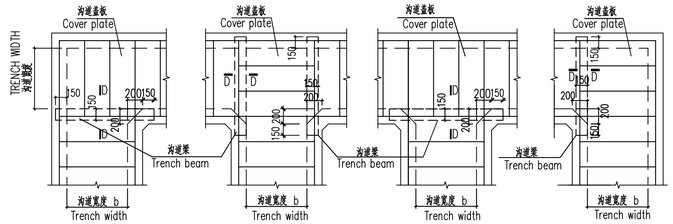


沟道与沟道交叉详图

INTERSECTION DETAIL OF TWO TRENCHES

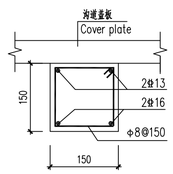


E-E



沟道交错布置图(室内)

Cover plate plan for intersection trench(lutdoor)



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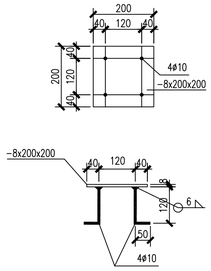
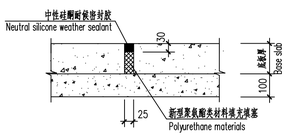
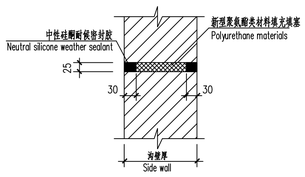


MJ-1



沟底伸缩缝示意图

Details of connection joint at base slab



沟壁伸缩缝详图

Details of connection joint at side wall

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 4 9 - 0 0 2 A

	姓名 Name	日期 Date
设计 Designed		2018.02.27
校核 Checked		2018.02.28
审核 Reviewed		2018.02.28
批准 Approved		2018.02.28
比例 Scale		

电缆沟详图(二)

Construction details of Cable Trench (Ⅱ)

图号 DWG NO. 30-B003701S-T0249-002

版本 REV. A

A1(841x594)

预埋件总说明

GENERAL DESCRIPTION OF THE EMBEDDED STEEL

1 材料：

1 Materials

1.1 预埋件应采用可焊性及塑性良好的钢材制作，锚筋不得采用冷加工钢筋。

1.1 The embedded metal parts are manufactured by use of steel with good weld ability and plasticity. Cold-worked steel shall not be used for anchor bars.

1.2 钢板、角钢和扁钢采用Q235B。

1.2 The structural carbon steel , grade Q235B, should be used for steel plate, angle steel and steel strip.

1.3 钢筋采用ASTMA615Grade40级钢筋（以Φ表示）及ASTMA615Grade60级（以⊕表示）。本图集中锚筋直径≤10用ASTM40级钢筋，锚筋直径>10用ASTMA615Grade60级钢筋。

1.3 Reinforced bars

Grade ASTM A615 Grade 40, plain bar (expression Φ),in the drawings used for anchor bars its dinette is less than 10mm.

Grade ASTM A615 Grade 60, the crescent threading bar (expression ⊕), used for anchor bars its diameter is not less than 10mm.

1.4 焊条、焊剂：当为手工电弧焊时，ASTMA615Grade40级钢筋用E43型，ASTMA615Grade60级钢筋用E50型，当为埋弧压力焊时，可采用E431型焊剂或其它性能相近的焊剂。

1.4 Electrodes and fluxes

For manual shielded metal are welding , the class E43 electrodes are used for grade I reinforced bars, and the class E50 electrodes are used for grade II reinforced bars,

For submerged are welding, the class E431 fluxes or equivalents may be used.

1.5 混凝土强度等级：预埋件应埋置于混凝土强度≥f'c=28MPa之构件内。

1.5 Concrete strength grade

The embedded parts shall be anchored in concrete member whose strength grade shall not be less than f'c=28MPa.

2 设计说明：

2. Description of design

2.1 受力预埋件，按承载能力极限状态计算。

2.1 The stressed embedded parts should be calculated according to the limit states design.

2.2 受拉、受剪预埋件，作用力均通过全部锚筋的重心。

2.2 For tension and shear embedded parts, the lines of action force shall pass through the center of all anchor bars.

2.3 受弯受剪预埋件，其锚筋对称于弯矩作用平面。

2.3 For flexural and shear embedded parts, the anchor bars shall be arranged in symmetry with moment acting-plan.

3 受拉钢筋的锚固

3 Development of deformed bars in tension

受拉钢筋的锚固长度取下表计算值和305mm的最大值。

The development length of bars in tension shall be the greater of values in the following table and 305mm.

受拉钢筋的锚固长度与直径之比 l_a/d_b

The ratio of development length to diameter l_a/d_b

应用情况 Condition	fy(MPa)	≤No.19 fc'(MPa)		≥No.22 fc'(MPa)	
		28	35	28	35
符合(a)或(b)情况 Conform to(a)or(b)	280	25	23	32	28
	420	38	34	47	42
其它情况 Other	280	38	34	47	42
	420	57	51	71	64

(a) 钢筋在延伸或搭接处的净间距≥d_b，净保护层≥d_b，且箍筋在l_a范围内不小于规范要求的最小规定。

(a) Clear spacing of bars or wires being developed or lap spliced not less than d_b, clear cover at least d_b, and stirrups or ties throughout l_a not less than the Code minimum.

(b) 钢筋在延伸或搭接处的净间距≥2d_b，净保护层≥d_b。

(b) Clear spacing of bars or wires being developed or lap spliced at least 2d_b and clear cover at least d_b.

(c) 当纵筋下面的新浇筑混凝土的厚度大于305mm时，表中值应乘以1.3。

(c) When there are more than 305mm of fresh concrete placed below horizontal reinforcement , the ratio in table should be multiplied by 1.3.

4 受拉钢筋的弯钩锚固

4 Development of standard hooks in tension

受拉钢筋的弯钩锚固长度取下表计算值、8d_b和153mm的最大值。

The development length of bars in tension shall be the greater of values in the following table, 8d_b and 153mm。

受拉钢筋的弯钩锚固长度与直径之比 l_{dh}/d_b

The ratio of development length fo standard hooks to diameter l_{dh}/d_b

应用情况 Cases	fy(MPa)	fc'(MPa)	
		28	35
符合(a)或(b)情况 Conform to(a)or(b)	280	10	9
	420	15	14
其它情况 Other	280	13	11
	420	19	17

(a) 对于≤No.32带90度弯钩的钢筋，符合(1)沿l_{dh}段内的垂直封闭箍筋或拉筋间距s ≤ 3d_b 或(2)沿l_{ext}段内的垂直封闭箍筋或拉筋间距s ≤ 3d_b。

(a) For 90-degree hooks of No. 32 and smaller bars

(1) enclosed along l_{dh} within ties or stirrups perpendicular to l_{dh} at s ≤ 3d_b or (2) enclosed along the bar extension beyond hook including the bend within ties or stirrups perpendicular to l_{ext} at s ≤ 3d_b.

(b) 对于≤No.32带180度弯钩的钢筋，符合沿l_{dh}段内的垂直封闭箍筋或拉筋间距s ≤ 3d_b。

(b) For 180-degree hooks of No. 32 and smaller bars enclosed along l_{dh} within ties or stirrups perpendicular to l_{dh} at s ≤ 3d_b.

5 预埋件的制作：

5 Fabrication of Embedded Part :

5.1 所有焊缝除注明者外均应满焊，并应满足钢筋焊接及验收规程JGJ 18-2003的要求。

5.1 All the welds shall be full welded (unless noted), and shall be conformed to JGJ18-2003《Code of Reinforcement Welding Practice and Acceptance》

5.2 预埋件的受力锚筋与锚板呈T型垂直焊接时，不得将锚筋弯成U型或L型后用角焊缝与锚板焊接，锚筋端部应采用埋弧压力焊或周边电弧焊与锚板焊牢，当锚筋直径d≤20mm时，宜优先采用埋弧压力焊。当d>20mm时，应尽量采用穿孔塞焊，穿孔塞焊的要求见下图，所有焊缝均应严格保证焊接质量并加强检查。

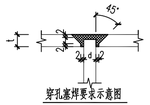
5.2 The anchor bar shall not be fillet welded with anchor plate as tee joint after the bar has been bent to "U" or "L" shape.

The end of anchor bar shall be submerged arc welding or shielded metal arc welding all around with anchor plate.

When the diameter of anchor bar d≤20mm, it shall be accepted submerged arc welding as possible; and when d>20mm, the plug welding shall be adopted as possible. The requirement of plug welding shall be conformed as shown in the following typical drawing. All the weldings shall be insure the quality strictly and inspected responsibly.

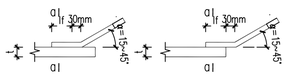
5.3 受拉锚筋（包括直锚筋及弯折锚筋）与锚板水平连接时，应采用双面角焊缝，焊缝长度l≥4d（Grade40级钢筋）或≥5d（HRB60级钢筋）。

5.3 Tension anchor bar , including straight bar and bent bar , shall be welded both sides with anchor plate for horizontal connection, length of weld l≥4d (bar Grade40) or ≥5d (bar Grade60)



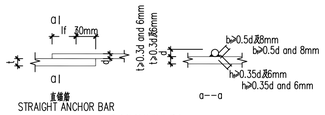
穿孔塞焊要求示意图

PLUG WELD DETAIL (TYP)



弯折锚筋

BENT ANCHOR BAR



直锚筋

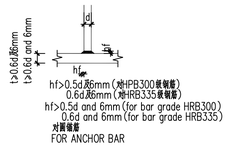
STRAIGHT ANCHOR BAR

水平受拉锚筋焊接要求

HORIZONTAL TENSION ANCHOR BAR (WELD REQ.)

5.4 对于锚筋或角钢锚筋采用T型手工焊时，焊缝的具体要求如下图所示：

5.4 The typical detail of manual welding for tee joint of anchor bar see the following drawing.



T型手工焊的要求

REQUIREMENT OF WELDING FOR TEE JOINT

5.5 预埋件各部尺寸要求制作准确，锚板尺寸宜采用负公差，以便于放入模板内。

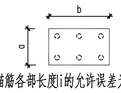
5.5 All the sizes of embedded part shall be fabricated accurately, The tolerance of anchor plate shall be negative for set into shuttering easily.

5.6 预埋件加工的允许误差：

5.6 Tolerance For Embedded Part Fabrication :

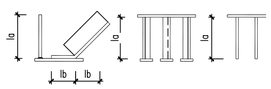
5.6.1 锚板边长a或b的允许误差为-2mm。

5.6.1 The tolerance for anchor plate edge a or b is -2mm。



5.6.2.锚筋各部长度li的允许误差为+10mm。

5.6.2 The tolerance for any length li of anchor bar is +10mm。



5.6.3 二面为整块锚板的预埋件，其锚筋长度l_a的允许误差为±3mm。

5.6.3 The tolerance for anchor bar ength l_a of double-faced anchor plate is±3mm。

5.6.4 圆锚筋的中心线偏差为±2mm。

5.6.4 The tolerance for center line of anchor bar is ±2mm。

5.6.5角钢锚筋重心线偏差为±2mm。

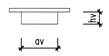
5.6.5 The tolerance for center line of angle is ±2mm。

5.6.6角钢锚筋重心线偏斜α为±2°。

5.6.6 The tolerance for angle α is ±2°。

5.6.7 抗剪钢板的宽度a_v和高度h_v的允许误差为+2mm。

5.6.7 The tolerance for width of shear bar a_v and high h_v is ±2mm。



5.6.8 锚筋对锚板面的垂直度偏差a应≤10mm及l_a/50。

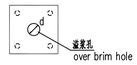
5.6.8 The tolerance for perpendicularly of anchor bar with the face of anchor plate shall be a ≤10mm and l_a/50。

5.6.9 溢浆孔直径d允许误差为±2mm。

5.6.9 The tolerance for diameter of over brim hole is ±2mm。

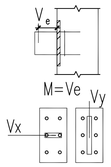
5.6.10 溢浆孔中心线偏差为±5mm。

5.6.10 The tolerance for center line of over brim hole is ±5mm。



5.6.11 直锚筋埋件中，x、y受力方向如下图所示。

5.6.11 The working direction is as follows for embeded parts with straight anchor bar.



Reference Drawings

Drwg. No.	Title

Revisions

Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

MLHVDC-LCS-ECI-DD-CS-0249-003A

	姓名 Name	日期 Date
设计 Designed		2018.02.27
校核 Checked		2018.02.28
审核 Reviewed		2018.02.28
批准 Approved		2018.02.28

预埋件总说明(一)

General Description of the Embeded Steel (I)

比例 Scale:

图号 DWG NO.: 30-B003701S-T0249-003

版本 REV.: A

A1(841x594)

6 施工说明：

6 Construction Instrution：

6.1 为保证预埋件与混凝土面的平整，建议在预埋件四角适当开小孔，以便钉子或螺丝与模板固定（适用于木模板）。

6.1 To make sure smoothly with the concrete surface, it is recommended to make small hole in each of four corners of embedded part for use nail and fix on the shuttering. (Applicable for shuttering board)

6.2 尺寸较大的预埋件位于构件混凝土浇灌面时，应在钢板中间开设排气孔，以保证混凝土浇灌密实。

6.2 For embedded part with large size as located at the concreting surface, the exhausted hole shal be made at the center of plate in security for the concrete densification.

6.3 当预埋件锚筋长度伸出构件表面时，允许根据构件尺寸弯折成虚线状态（见图1）。

6.3 For the length of embedded bar extends out the surface of member, the bar bent as the dotted line shown in the following detail 1 is permit bar bent as the dotted line shown in the following detail 1 is permit.

6.4 板内受压预埋件之锚筋，允许根据板厚弯折或拉成虚线状态（见图2）。

6.4 For the anchor bar of compression embedded part in slab, the bar may be bent or pulled as dotted line shown in the detail 2.

6.5 角钢及扁钢预埋件之弯折锚筋，在预埋件放入构件模板后弯折成如图3中虚线状态。锚筋按等间距通长放置者须与构件中之钢筋相配合。

6.5 The anchor bar for angle and plate of embedded part shall be bent until be located into the shuttering as shown dotted line in detail 3. The equal spacing anchor bars shall cooperated with the reinforcements in member.

6.6 预埋件的锚筋凡未注明间距者，为均匀排列。

6.6 The anchor bars of embedded part shall be arranged in equal spacing for it has not noted.

6.7 预埋件各部尺寸要求制作准确。如果预埋件钢板尺寸与构件模板尺寸相同时，则在下料时钢板尺寸应用负公差。以便入模。

6.7 All sizes of embedded part shall be fabricated accurately. If the size of embedded plate is same as the member shuttering, it shall negative tolerance for set into shuttering easily.

6.8 构件或工艺安装完毕后，预埋件外露部分涂防腐涂料二度。

6.8 After erection of civil member or technical facility, it shall be painted with one primer coal of red paint and two coats of surface paint on the exterior surface of embedded part.

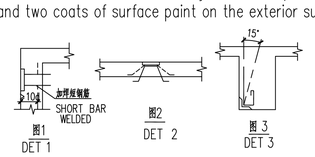


图1 DET 1　　图2 DET 2　　图3 DET 3

7 预埋件编号说明：

7 Explanation of Embedded Part Mark

7.1 钢板预埋件编号：

7.1 Mark of Embedded Steel Plate：

M－表示机务专业所设置的预埋件；

M – embedded part for mechanical discipline；

E－表示电气专业所设置的预埋件；

E – embedded part for electric discipline；

C－表示土建专业所设置的预埋件。

C – embedded part for civil discipline；

A、B、C、D、E－表示预埋件的荷载级别，可按锚筋的不同直径区分。

A、B、C、D、E – loading class, it may be distinguished by the diameter of anchor bar.

1、2－表示锚筋的锚固长度不同。

1、2 – the distinction of embedment length of anchor bar.

预埋件编号中数字代表规格及埋入方位见图4。例如：

The number in the mark of embedded part denotes the size and embedded direction, see detail 4, for example:

钢板预埋件：M2030B1。其中"M"表示机务预埋件，前一组数字"20"表示钢板边长为200毫米而且表示沿构件纵方向设置。后一组数字"30"表示钢板边长为300毫米，而且表示沿构件横方向设置。"B"表示荷载级别。M3020B1可套用M2030B1，但埋设方向和承载能力不同。所谓构件纵方向，对梁来说是梁的长度方向，对柱是高度方向，对板是跨度方向（槽形板的肋按梁考虑，面板按板考虑）。在难于区分的特殊情况下，宜注明尺寸。

Plate embedded part：M2030B1, where

"M" – embedded part for mechanical discipline.

"20" – edge length 200mm of steel plate located along longitudinal direction of member;

"30" – edge length 300mm of steel plate located along transversal direction of member;

"B" – load class;

"1" – length kind of embedded bar;

M2030B1 is applicable for M3020B1, only the difference direction.

The longitudinal direction of member means longitudinal direction of beam, height direction of column or direction of slab span(the rib of channel slab is regarded as a beam and faced slab regarded as a slab). The size shall be noted on for the special case that is distinguished difficult.

7.2 扁钢预埋件E－100(50)A1。其中"－100(50)"表示宽为100毫米，长为500毫米的扁钢预埋件。E－100(※)A1，其中"(※)"表示预埋件长度。单位用厘米，在施工图上注明。

7.2 Embedded Strip Steel

E－100(50)A1, where

－100(50) – denotes width 100mm, length 500mm for embedded strip steel.

E－100(※)A1, where (※) denotes lenth (cm) of embedded part.

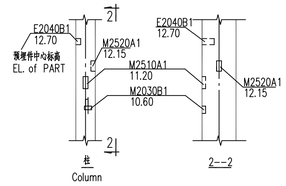


图4 预埋件位置标高编号标注法

DET 4 LOCATION, EL and MARK OF EMBEDDED PART (TYP.)

7.3 角钢预埋件CL63(30)A1，其中"L63(30)"表示L63x6长为300毫米的角钢预埋件。"A"为荷载级别"L"表示仅用于梁侧底的预埋件。符号(※)意义同扁钢预埋件。

7.3 Embedded Angle Steel

For example: CL63(30)A1, where "L63(30)" – embedded angle L63x6, length 300mm.

"A" – load class;

"L" – embedded part located in the bottom of beam lateral side.

"Z" – embedded part in column and other than the bottom of beam lateral side.

(※) – length (cm) of embedded part, noted on detail.

7.4 两个及两个以上的相同预埋件且埋设方位一致又紧靠在一起拼接埋设时的表示方法见图5。如埋设方位一致但不紧靠在一起者表示方法见图6。

7.4 Two or more same embedded parts and same embedded direction located together (or not) side by side be marked as detail 5 or 6.

7.5 除特殊注明外，埋件标高均为锚板中心标高。

7.5 The elevation marked is the central elevation of embeded parts otherwise noted specailly.

7.6 非标准的预埋件可由同级别的标准预埋件拼成。

7.6 Non-standard embeded part can be made of standard parts with same grade.

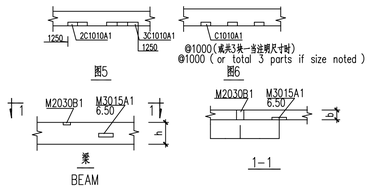
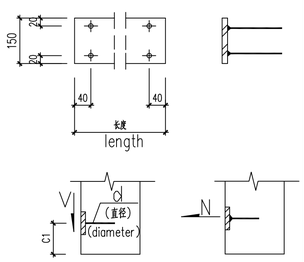


图5　　图6

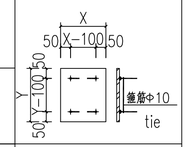
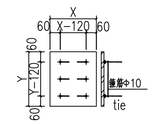
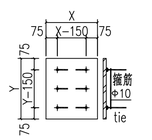
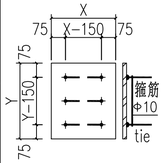
扁钢预埋件

EMEDDED STRIP STEEL

埋件名称 NO.	扁钢(mm) strip steel		锚筋(mm) anchor bar			备注 remarks
	规格 size	长度 length	直径 diameter	根数 number	长度 length	
－150(※)A1	－150X12	通长 whole	13@120		150	

直锚筋钢板预埋件

EMBEDDED STEEL PLATE WITH STRAIGHT ANCHOR BAR

埋件编号 NO.		锚板 steel plate (mm) 宽 width	长 length	厚 thickness	锚筋 anchor bar (mm) 数 number	直径 diameter	长 length	焊缝 weld size (mm)	简图 sketch
M E1010 C	A1	100	100	8	4	10	180	6	
M E1015 C	A1	100	150	8	4	13	180	6	
M E1515 C	A1	150	150	8	4	13	180	8	
M E1530 C	A1	150	300	8	6	13	180	8	
M E2025 C	A2	200	250	10	4	13	468	8	
	B2	200	250	12	4	16	624	10	
M E2030 C	A2	200	300	12	6	13	468	8	
M E2040 C	A2	200	400	12	6	13	468	8	
M E2525 C	A1	250	250	8	4	13	180	8	
M E3030 C	A1	300	300	8	4	13	180	8	
	B2	300	300	14	4	16	624	12	

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

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拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE　施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 4 9 - 0 0 4 A

	姓名 Name	日期 Date
设计 Designed		2018.02.27
校核 Checked		2018.02.28
审核 Reviewed		2018.02.28
批准 Approved		2018.02.28

预埋件总说明(二)

General Description of the Embeded Steel (Ⅱ)

比例 Scale　图号 DWG NO. 30-B003701S-T0249-004　版本 REV. A

A1(841x594)

总说明
GENERAL DESCRIPTION

1 一般说明及适用范围

1, General description and limitation:

1.1 本图集为悬挂运输设备轨道（简称轨道）与一般混凝土梁或钢梁连接的施工图。

1.1, This drawing descripts the connection of suspended transportation equipment and general concrete beam or steel beam.

1.2 本图集适用于悬挂在混凝土梁或钢梁上的一台机构工作级别为M1~M5与手动葫芦配套的手动单轨小车、电动葫芦或一台起重机工作级别为A1~A5的手动梁式悬挂起重机、电动单梁悬挂起重机。

1.2, This drawing is suitable for monorail trolley and electric hoist whose working class is M1~M5 with manual hoist suspended on concrete or steel beam It is also suitable for manual beam suspension crane and electric single beam suspension crane of working class A1~A5.

1.3 本图集直线轨道适用于额定起重量≤10t的手动、电动葫芦及额定起重量≤25t的电动单梁悬挂起重机，轨道计算跨度为2.5~6m。

1.3, The straight rail is suitable for manual or electric hoist whose capacity no more than 10t and those electric single beam suspended crane whose capacity no more than 25t. The calculation span is 2.5~6.0m。

1.4 本图集弧线段轨道适用于额定起重量≤10t的手动、电动葫芦，轨道支承点夹角为45°,轨道曲率半径2~4m,

1.4, The arc rail is suitable for manual or electric hoist whose capacity no more than 10t. The included angle of the connection is 45°. Curvature radius of the rail is 2~4m.

1.5 本图集构件设计使用年限为50年，安全等级为二级。

1.5, Design service life of the component is 50 years. Safety class is Class Ⅱ.

2 设计依据

2, The design basis of the main specifications,standards

现行国家规范及标准：

《建筑结构荷载规范》GB 50009-2012

《Load code for the design of building structures》(GB 50009-2012)

《混凝土结构设计规范》GB 50010-2010(2015年版)

《Code for design of concrete structures》(GB 50010-2010)(2015)

《钢结构设计规范》GB 50017-2003

《Code for design of steel structures》(GB 50017-2003)

《钢结构工程施工质量验收规范》GB 50025-2001

《Code for acceptance of constructional quality of steel structures》(GB 50025-2001)

《钢筋焊接及验收规程》JGJ 18-2003

《Specification for welding and acceptance of reinforcing steel bars》(JGJ 18-2012)

《悬挂运输设备轨道设计计算》（国标图集06CG08）

《Calculation of suspension rail transportation equipment design》(GB Atlas 06CG08)

3 材料

3, Material:

3.1 结构材料

3.1, Structural material:

3.1.1 轨道工字钢选自《热轧工字钢尺寸、外形、重量及允许偏差》GB/T706-1988，由Q235-B或Q345-B钢轧制而成。

3.1.1, I-steel rail type shall be chosen from《Hot rolled beam steel - Dimensions, shape, weight and tolerances》GB/T706-1988 The I-steel rail is rolled by Q253-B or 345-B.

3.1.2 其他材料如钢板、角钢等采用Q235-B钢材。

3.1.2, Other materials such as steel plate and angle steel shall use Q235-B.

3.1.3 受拉连接螺栓采用摩擦型大六角高强螺栓10.9级，配双螺母。

3.1.3, The tension connecting bolt shall use friction type high strength 10.9 level bolt with six angle, using double nuts.

3.1.4 方斜垫圈采用《工字钢用方斜垫圈》GB/T852-1988

3.1.4, Square sloping washer shall be chosen from《Square taper washers for I section》GB/T852-1988.

3.1.5 Q235-B轨道拼接采用低氢型焊条E43系列，Q345-B轨道拼接采用低氢型焊条E50系列；其他构件焊接采用非低氢型焊条E43系列。

3.1.5, Low-hydrogen electrode E43 series shall be used for Q235-B rail and Low-hydrogen electrode E50 series for Q345-B rail. non-low-hydrogen electrode E43 series should be used for other components.

3.2 油漆：参照各具体工程。与电动葫芦滚动轮接触的工字钢表面不油漆。

3.2, Paint sees each specific project. The surface of the I-steel rail rolling conctacted with the electric hoist should not be painted.

4 其他

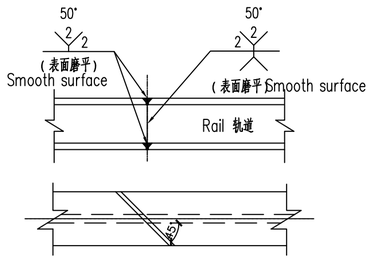
4, Others

4.1 轨道接头拼接位置宜设在距支撑点1/3~1/4跨度范围内，采用等强度透焊焊接，见详图，焊缝质量等级为一级。

4.1, The joint connection shall be located at the 1/3~1/4 span range from the support point. The equal-strength weld should be used at the joint connction, see details. Weld quality level shall be the first grade.

4.2 所有连接件角焊缝外观质量标准应符合二级。

4.2, The visual inspection of the fillet weld for all connecting component shall reache the second grade.



拼接焊缝详图

Butt weld detail

4.3 混凝土支承梁两侧成对埋件，可加工成组合埋件，也允许锚筋采用单面搭接焊接10d。

4.3, The pair of the lateral embedments on both sides of one concrete beam could be made as a combanation embedment. The anchor bar of the embedment could be single lap welding 10d.

5 本图集选用注意事项：

5, Selecting attention

5.1 轨道应在工艺专业要求的轨道型号范围内选用。

5.1, The rail type should be included in the requirement of the technical specialty.

5.2 弧线段轨道两端均要求必须至少连续一跨直线段轨道。

5.2, Both ends of the arc rail shall be connected with at least one span of straight rail.

5.3 钢支承梁下翼缘厚度不小于连接件顶板的厚度，下翼缘宽度也应满足连接要求。

5.3, The bottom flange thickness of the supporting steel beam should be no less than the thickness of the roof joint plate. The bottom flange width of the supporting steel beam should also meet the requirement of the conncetion.

5.4 支承梁底至轨道顶之间的距离hd，应满足连接件的最小尺寸要求，见下表：

5.4, The distance between the bottom of the supporting beam and the top of the rail is hd which should satisfy the minimum size of the connecting pieces. The requirement is as follows:

Capacity 起重量	1t		2t		3t		5t		10t		15t		25t	
Beam Type 支承梁类别	Concrete 混凝土	Steel 钢	Concrete 混凝土	Steel 钢	Concrete 混凝土	Steel 钢	Concrete 混凝土	Steel 钢	Concrete 混凝土	Steel 钢	Concrete 混凝土	Steel 钢	Concrete 混凝土	Steel 钢
Minimum hd hd最小尺寸	150	150	150	150	150	150	200	200	200	200	250	250	330	250

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: 施工图 Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 4 9 - 0 0 5 A

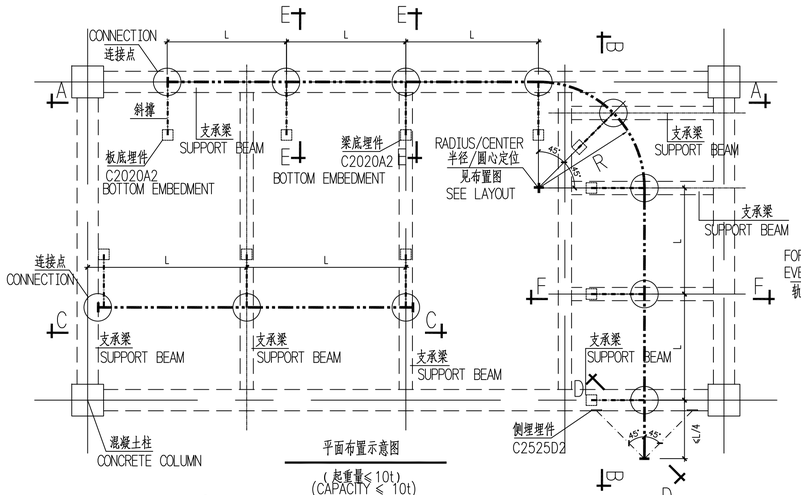
	姓名 Name	日期 Date
设计 Designed		2018.02.27
校核 Checked		2018.02.28
审核 Reviewed		2018.02.28
批准 Approved		2018.02.28

单轨吊总说明

GENERAL DESCRIPTION OF MOTOR HOIST MONORAIL

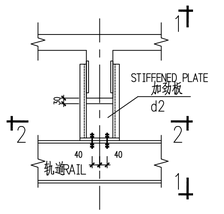
比例 Scale | 图号 DWG NO. 30-B003701S-T0249-005 | 版本 REV. A

A2(594x420)



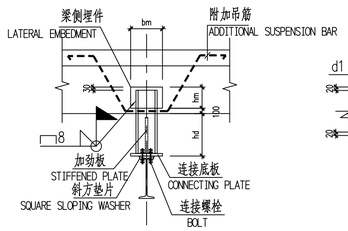
平面布置示意图

(起重量≤10t)
(CAPACITY ≤ 10t)
SCHEMATIC DIAGRAM OF PLANE LAYOUT



轨道与支承梁垂直连接立面图

CONNECTION ELEVATION
WITH VERTICAL SUPPORT BEAM



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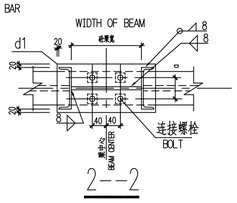

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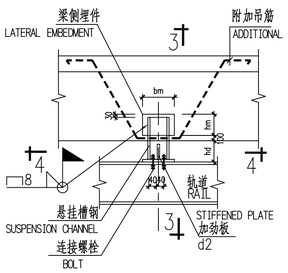
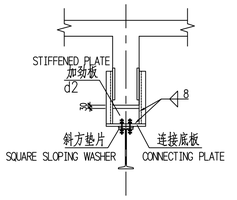


轨道与支承梁平行连接立面图

CONNECTION ELEVATION
WITH PARALLEL SUPPORT BEAM



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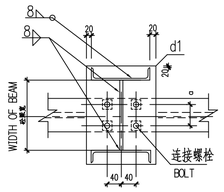

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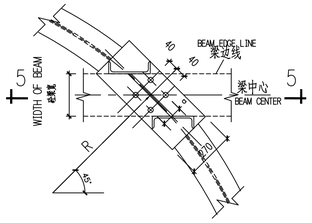
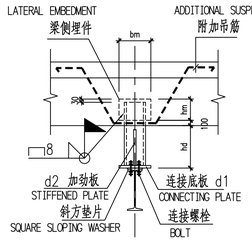
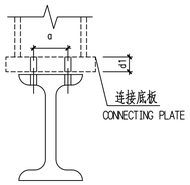


轨道与支承梁45°连接平面图

CONNECTION PLAN
WITH 45° SUPPORT BEAM



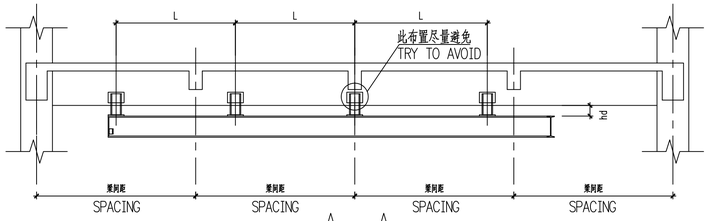
5--5



工字钢轨道翼缘钉距规线表及连接底板厚度d1加劲板厚度d2

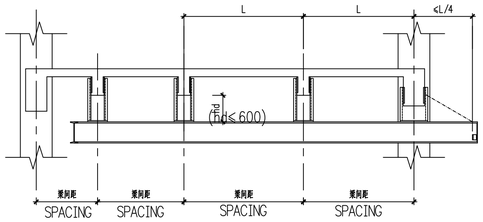
I-STEEL RAIL GUAGE a, THICKNESS d1 OF CONNECTING PLATE, THICKNESS d2 OF STIFFENED PLATE

I-STEEL RAIL 工字钢轨道	I20a	I22a	I25a	I28a	I32a	I36a	I40c	I45c	I50c	I56c	I63c
a (mm) a值 (mm)	54	54	64	64	70	74	80	84	94	104	110
MAX. HOLE DIAMETER ALLOWED ON THE FLAGE 允许最大孔径	17	19	21.5	21.5	21.5	23.5	23.5	25.5	25.5	25.5	25.5
THIOKNESS d1 OF CONNECTING PLATE 连接底板d1厚度	20	20	20	20	22	22	22	22	25	25	25
THIOKNESS d2 OF STIFFENED PLATE 加劲板d2厚度	10	10	10	10	12	12	16	16	18	18	18

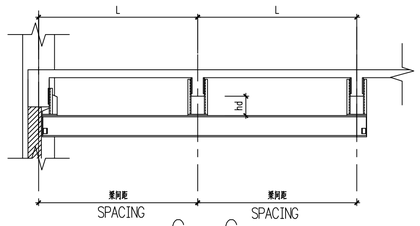


A--A

(hd≤600)



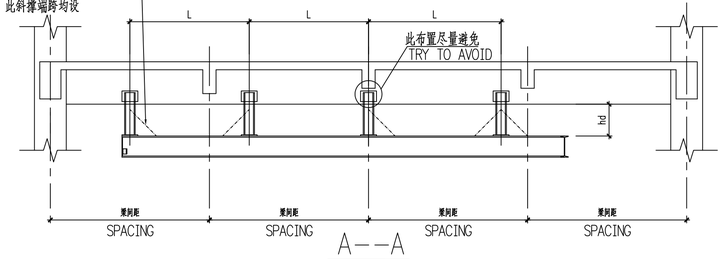
B--B



C--C

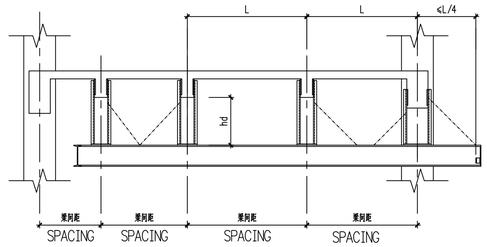
(hd≤600)

FOR END CONNECTION AND
EVERY 36m STRAIGHT LINE
轨道直段每≥36m时需增设
此斜撑端跨均设



A--A

(600< hd≤2500)



B--B

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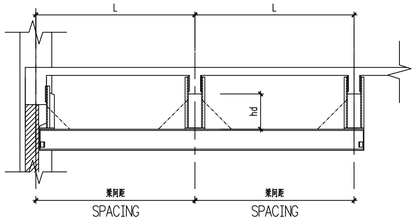
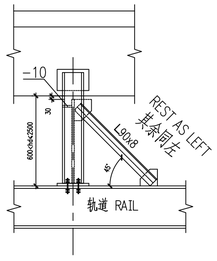


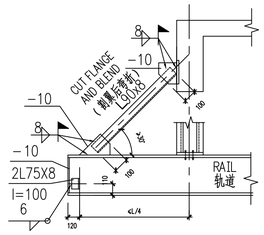
C--C

(600< hd≤2500)
BOTTOM EMBEDMENT

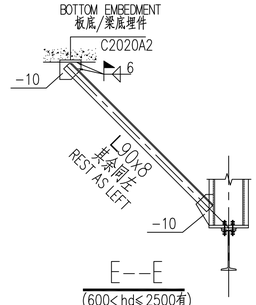


轨道与支承梁平行情况斜撑连接图

BRACING CONNECTION DETAILS
FOR RAIL PARALLEL WITH SUPPORT BEAM

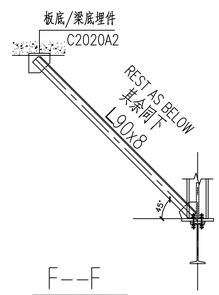


D--D



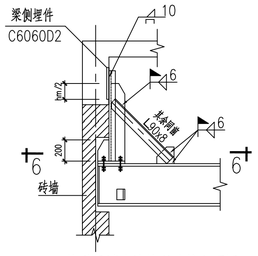
E--E

(600< hd≤2500有)
(FOR 600<hd≤2500)



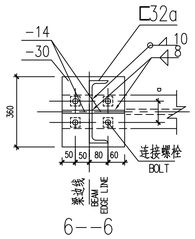
F--F

(600< hd≤2500有)
(FOR 600<hd≤2500)

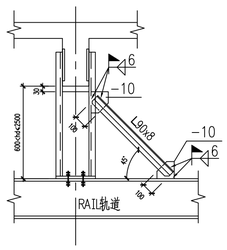


轨道尽端有墙情况连接立面图

(600< hd≤2500有斜撑)
END CONNECTION DETAILS
IF WALL AT THE END
(BRACING NEEDED FOR 600<hd≤2500)

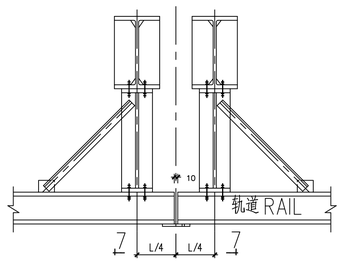


6--6



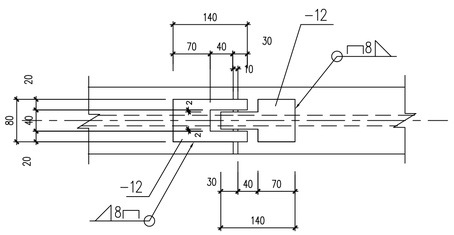
轨道与支承梁垂直情况斜撑连接图

BRACING CONNECTION DETAILS
FOR RAIL PERPENDICULAR TO SUPPORT BEAM



伸缩缝处立面详图

DETAIL AT EXPANSION JIONT



7--7

Reference Drawings

Drwg. No.	Title

Revisions

	Name	Date	Notes

巴基斯坦默蒂亚里—拉合尔±660kV直流输电工程

±660kV HVDC Project from Matiari to Lahore in Pakistan

NATIONAL TRANSMISSION & DESPATCH COMPANY (NTDC) LIMITED PAKISTAN

CONTRACTOR 国家电网 STATE GRID 中国电力技术装备有限公司 CHINA ELECTRIC POWER EQUIPMENT AND TECHNOLOGY CO.,LTD.

SUBCONTRACTOR East China Electric Power Design Institute Co.,Ltd.

拉合尔±660kV换流站

±660kV Lahore Converter Station

设计阶段 PHASE: Detail Design

M L H V D C - L C S - E C I - D D - C S - 0 2 4 9 - 0 0 6 A

	姓名 Name	日期 Date
设计 Designed		2018.02.27
校核 Checked		2018.02.28
审核 Reviewed		2018.02.28
批准 Approved		2018.02.28
比例 Scale		图号 DWG NO.

混凝土支承梁与轨道连接图

MONORAIL CONNECTION DETAILS WITH CONCRETE BEAM

30-B003701S-T0249-006

版本 REV. A

A1(841x594)